

Zen Nguyen

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TECHNICAL SKILLS

- **Languages:** C, C++, Go, Python
- **Backend & Systems:** gRPC, TCP Sockets, Raft Consensus Algorithm, RESTful APIs
- **Infrastructure & DB:** Linux (WSL), Docker, PostgreSQL

TECHNICAL PROJECTS

TitanSwarm (Autonomous AI Co-Pilot) & TitanStore (Raft Database) Jan 2026 - Present
Personal Project · Python, Go, LangChain, FAISS

- Ensured high-availability state management by engineering TitanStore, a **distributed computing** memory bank in Go that implements the Raft consensus protocol and a custom Write-Ahead Log (WAL) to guarantee fault tolerance.
- Enabled high-throughput data synchronization by designing a custom TCP client API that facilitates real-time status updates between Python workers and the Go database cluster, supporting scaling to 100+ users.

Gridlock Casino (2D Arcade Engine) Feb 2026 - Present
Collaborative Project · Java, Swing, Maven, JUnit

- Optimized rendering performance by architecting a 2D grid-based engine in Java that manages 60-FPS cycles and concurrent user input within a strict MVC architecture.
- Validated core engine logic and **multithreaded programming** stability by managing build automation with Apache Maven and executing comprehensive JUnit testing across a 6-developer team.

SFU Course Tracker (sfucourseplanner.me) Nov 2025 - Jan 2026
Personal Project · Python, Docker, AWS, FastAPI

- Reduced infrastructure overhead by migrating the platform to AWS (EC2) and implementing **Docker** container orchestration for streamlined deployment cycles.
- Increased data throughput by 10x by engineering an asynchronous scraping pipeline with semaphore-based rate limiting, ensuring efficient interaction with the **networking software stack**.

EDUCATION

Bachelor of Science, Computing Science May 2025 - Present
Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 - Apr 2025

WORK EXPERIENCE

Server · Pho Goodness Restaurant Jan 2024 - Present
Burnaby, BC

- Streamlined high-pressure operations by coordinating effectively with a diverse team to manage service for 100+ guests per shift, demonstrating the ability to thrive in a **collaborative environment**.
- Maintained academic excellence with a 3.74 CGPA while balancing 20+ hours of weekly service, showcasing the discipline required for complex technical project delivery.